

MATH 161A - Applied Probability & Statistics

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1. Skipped

2. Probability

2.1 Sample Spaces and Events

Probability

- Probability is used as a mathematical tool to understand or describe random phenomena, chance variation, and uncertainty.
- We will use probability as a tool to evaluate the reliability of our conclusions about the population when we only have sample information.
- Probability is used in a variety of fields including genetics, quality control, and finance.
- Probability is used as a model for situation for which outcomes occur randomly.
- Such situations are called **experiments**.
- Examples:
 - Toss a coin.
 - Roll a die.
 - Draw a card from a deck.

Vocabulary

- An action or process whose outcome is subject to uncertainty is called an **experiment**.
- The possible outcomes of an experiment are called **sample space**.
- An **event** is a specific outcome or a set of outcomes from an experiment.
- A **simple event** is an event that consists of a single outcome from sample space S .

Example What is the outcome of rolling an even (E), rolling greater than 3 (F), and rolling a 2 (G) on a 6-sided die?

$$S = 1, 2, 3, 4, 5, 6$$

$$E = 2, 4, 6$$

$$F = 4, 5, 6$$

$$G = 2$$

A Note on Venn Diagrams All Venn Diagrams in this course require a rectangle to represent the sample space. Events are represented by circles within the rectangle.

Operations

Set Operations

- $A \subseteq B$: A is a subset of B .
- $A \cap B$: the intersection of A and B (elements in both A and B).
- $A \cup B$: the union of A and B (elements in either A or B).
- A^c or A' : the complement of A (elements in S that are not in A).

Logical Operations

- A or B : $A \vee B$
- A xor B : $A \oplus B$
- A and B : $A \wedge B$
- not A : $\neg A$

Mutually Exclusive Events

- Events A and B are **mutually exclusive** if they cannot both occur at the same time.
 - In terms of set operations: $A \cap B = \emptyset$.
 - Example: Rolling a die and getting an even number (E) or an odd number (O) are mutually exclusive events.
-

Identities

$$A \cup A' = S$$

$$A \cap A' = \emptyset$$

DeMorgan's Laws

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

$$\neg(A \vee B) = \neg A \wedge \neg B$$

$$\neg(A \wedge B) = \neg A \vee \neg B$$

2.2 Probability

Definition Probability is a function that maps the sample space S to $\mathbb{R}$. The sample space is the **domain** and the real numbers are the **codomain**.

Axioms of Probability

1. Non-negativity: For any event A , $P(A) \geq 0$.
2. Normalization: $P(S) = 1$, where S is the sample space.
3. Additivity: For any two mutually exclusive events A and B , $P(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_{i=1}^n P(A_i)$.

$$P(\cup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$$

Not axioms but useful observations

1. $P(A) < 1$ for any event A .
2. $P(A^c) = 1 - P(A)$.
3. $P(A \cap B) = P(A) + P(B) - P(A \cup B)$ for any two events A and B .

Proving Observations

Observation 1: $P(\emptyset) = 0$

$$\begin{aligned} P(\emptyset) &= 0 \\ \emptyset \cup S &= S \\ \emptyset \cap S &= \emptyset \\ P(\emptyset \cup S) &= P(S) \\ P(\emptyset) + P(S) &= P(S) \quad (\text{by axiom 3}) \end{aligned}$$

$\emptyset$ and S are mutually exclusive.
Therefore, $P(\emptyset) = 0$ by algebra.

Observation 2: $P(A^c) = 1 - P(A)$

$$S = A \cup A^c$$

$$\emptyset = A \cap A^c$$

$$P(A^c \cup A) = P(S)$$

$$P(A^c) + P(A) = P(S) \quad (\text{by axiom 3})$$

$$P(A^c) + P(A) = 1$$

$$P(A^c) = 1 - P(A)$$

Observation 3: $P(A \cup B) = P(A) + P(B)$ if $A \cap B = \emptyset$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \quad \text{for mutually exclusive } A_i$$

Let $A_1 = A$, $A_2 = B$, $A_3 = A_4 = \emptyset$

$$P(A \cup B \cup \emptyset \cup \emptyset) = P(A) + P(B) + 0$$

$$P(A \cup B) = P(A) + P(B) \quad (\text{by axiom 3})$$

Identities for the next proof

$$P(A) = P(A \cap B') + P(A \cap B)$$

$$P(B) = P(A' \cap B) + P(A \cap B)$$

$$P(A \cup B) = P(A \cap B') + P(A \cap B) + P(A' \cap B)$$

$$1 = P(A \cap B') + P(A \cap B) + P(A' \cap B) + P(A' \cap B')$$

Observation 4: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$P(A \cup B) = P(A \cap B') + P(A \cap B) + P(A' \cap B)$$

$$= P(A) + P(A' \cap B)$$

$$= P(A) + [P(B) - P(A \cap B)]$$

$$= P(A) + P(B) - P(A \cap B)$$

Other Propositions

$$A \subseteq B \implies P(A) \leq P(B)$$

$$B = (A \cap B) \cup (A' \cap B)$$

$$A \cap B = A$$

$$B = A \cup (A' \cap B)$$

$$P(B) = P(A) + P(A' \cap B) \quad (\text{by axiom 3})$$

$$P(B) \geq P(A) \quad \text{since } P(A' \cap B) \geq 0 \quad (\text{by axiom 1})$$

For any A , $A \subseteq S$

$$P(A) \leq P(S)$$

$$\therefore P(A) < 1$$

Example If $P(C) = .46$, $P(D) = .47$, $P(C \cup D) = .62$, find the following:

$$P(C') = 1 - P(C) = 1 - .46 = .54$$

$$P(C \cap D) = P(C) + P(D) - P(C \cup D) = .46 + .47 - .62 = .31$$

$$P(C \cap D') = P(C) - P(C \cap D) = .46 - .31 = .15$$

$$P(C' \cup D') = 1 - P(C \cup D) = 1 - .62 = .38$$

Probability Square Trick

Example There are 5 simple events in a sample space. $S = E_1, E_2, E_3, E_4, E_5$ with probabilities p_1, p_2, p_3, p_4, p_5 . Suppose $p_3 = 2p_2 = 2p_4$ and $p_2 = 2p_1 = 2p_5$. Find p_1, p_2, p_3, p_4, p_5 .

$$p_1 + p_2 + p_3 + p_4 + p_5 = 1$$

$$p_3 = 2p_2$$

$$p_3 = 2p_4$$

$$p_4 = p_2$$

$$p_2 = 2p_1$$

$$p_2 = 2p_5$$

$$p_1 = p_5$$

Solve for the 5 unknowns with the 5 equations:

2.3 Counting Techniques

Equally Likely Outcomes

If all outcomes in a sample space are equally likely for any event A , then

$$P(A) = \frac{\text{Number of outcomes in } A}{\text{Number of outcomes in } S} = \frac{n(A)}{n(S)}$$

where $n(A)$ is the number of outcomes in event A and $n(S)$ is the number of outcomes in sample space S .

Combinatorics

Combinatorics is the branch of mathematics dealing with combinations of objects in specific sets under certain constraints. It provides tools for counting and arranging objects. Some basic combinatorial concepts include:

- **Permutations:** The number of ways to arrange a set of objects in a specific order.
- **Combinations:** The number of ways to select a subset of objects from a larger set, without regard to the order of selection.

Counting Subsets Let $\Omega = a, b, c, d, e, f, g, h, i, j$. How would we find all possible subsets of Ω ?

$$Num = 2^{\text{len}(\Omega)} = 2^{10} = 1024$$

This is because each element can either be included or excluded (2 options) from a subset, leading to 2^n possible subsets for a set with n elements.

Permutations and Combinations

Permutations A permutation is an arrangement of objects in a specific order. The number of permutations of k objects chosen from n distinct objects is given by the following formula:

$$P_{k,n} = \frac{n!}{(n-k)!} = n \times (n-1) \times (n-2) \times \dots \times (n-k+1)$$

Where $P_{k,n}$ is the number of permutations of n objects taken k at a time, $n!$ is the factorial of n , and $(n-k)!$ is the factorial of $(n-k)$.

Example: Suppose we want to rank 5 candidates Amy, Bai, Catherine, David, and Eduardo in preference for a position. How many different ways can we rank (1.) all of the candidates and (2.) only 2 of the candidates?

$$\begin{aligned} 1. \quad P_{5,5} &= \frac{5!}{5-5!} = \frac{5!}{1} = 120 \\ 2. \quad P_{5,2} &= \frac{5!}{5-2!} = \frac{5!}{3!} = 20 \end{aligned}$$

Combinations A combination is a selection of objects without regard to the order of selection. The number of combinations of k objects chosen from n distinct objects is given by the following formula:

$$C_{k,n} = \frac{n!}{k!(n-k)!} = \frac{P_{k,n}}{k!}$$

Where $C_{k,n}$ is the number of combinations of n objects taken k at a time, $n!$ is the factorial of n , $k!$ is the factorial of k , and $(n-k)!$ is the factorial of $(n-k)$.

Example: How many hands of 3 cards can be drawn from a standard deck of 52 playing cards?

$$C_{3,52} = \binom{52}{3} = \frac{52!}{3!(52-3)!} = \frac{52!}{3!49!} = 22100$$

Example: A playlist contains 100 songs. 10 of these songs are by The Strokes. We can find the probability of the first song by The Strokes being the 5th song played by finding the ratio of the number of favorable outcomes to the total number of outcomes.

$$\begin{aligned}
 P(\text{1st Strokes on 5th}) &= \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} \\
 &= \frac{P_{4,90} \cdot P_{1,10}}{P_{5,100}} = \frac{\frac{90!}{(90-4)!} \cdot 10}{\frac{100!}{(100-5)!}} \\
 &= .0679
 \end{aligned}$$

Another way of thinking about this would be that there are $\binom{100}{10}$ different ways to choose the 10 Strokes songs. Now if we choose 9 of the last 95 songs to be Strokes songs, there are $\binom{95}{9}$ ways to do this, leaving 4 non-Strokes songs and 1 Strokes song in the first 5 songs. The probability is then:

$$P(\text{1st Strokes on 5th}) = \frac{\binom{95}{9}}{\binom{100}{10}} = .0679$$

Example: A university warehouse has recieved 25 printers, of which 10 are laser printers and 15 are inkjet printers. If 6 printers are randomly selected for testing, what is the probability that exactly 3 of those selected are laser printers?

Let D_3 be the event that exactly 3 of the selected printers are laser printers.

$$\begin{aligned}
 P(D_3) &= \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} \\
 &= \frac{N(D_3)}{N} = \frac{\binom{15}{3} \cdot \binom{10}{3}}{\binom{25}{6}} = .3083
 \end{aligned}$$

2.4 Conditional Probability

A conditional probability of an event is the probability that the event occurs given that some other event has already occurred. Conditional probabilities can prove to be counterintuitive and may lead to surprising results.

Definition: The conditional probability of an event A given that an event B has occurred is denoted by $P(A|B)$ and is defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Example: If you flip 3 coins, what is the probability that exactly 2 are heads (B)?

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

$$B = \{HHT, HTH, THH\}$$

$$A = \{HHH, HTT, HTH, HHT\}$$

$$\begin{aligned} P(A \cap B) &= \frac{2}{8} \\ P(B) &= \frac{3}{8} \\ P(A|B) &= \frac{P(A \cap B)}{P(B)} = \frac{\frac{2}{8}}{\frac{3}{8}} = \frac{2}{3} \end{aligned}$$

Multiplication Rule

The multiplication rule is a way to find the probability of the intersection of two events. It states that:

$$P(A \cap B) = P(A|B) \cdot P(B)$$

$$P(A \cap B) = P(B|A) \cdot P(A)$$

CHECK THIS FOR ACCURACY

General Multiplication Rule: For any number of events $A_1, A_2, \dots, A_n$:

$$P\left(\bigcap_{i=1}^n A_i\right) = P(A_1) \cdot P(A_2|A_1) \cdot P(A_3|A_1 \cap A_2) \cdots P(A_n|A_1 \cap A_2 \cap \cdots \cap A_{n-1})$$

Law of Total Probability

For any events A and B , we can get the following from the multiplication rule:

$$\begin{aligned}P(A \cap B) &= P(A|B)P(B) \\P(A \cap B') &= P(A|B')P(B')\end{aligned}$$

: Note that B and B' form a *partition* of Ω .

General Law of Total Probability: Let $B_1, \dots, B_n$ be a partition of the sample space Ω . Then for any event A :

$$P(A) = \sum_{i=1}^n P(A|B_i)P(B_i)$$

Example: Suppose that we have 2 boxes with marbles in them. Box 1 has 1 white and 3 black marbles. Box 2 has 2 white and 2 black marbles. We randomly select a box and then randomly select a marble from that box. What is the probability that we select a black marble?

$$\begin{aligned}P(\text{Black}) &= P(\text{Black}|\text{Box1})P(\text{Box1}) + P(\text{Black}|\text{Box2})P(\text{Box2}) \\&= \left(\frac{3}{4}\right)\left(\frac{1}{2}\right) + \left(\frac{2}{4}\right)\left(\frac{1}{2}\right) \\&= \frac{3}{8} + \frac{2}{8} = \frac{5}{8}\end{aligned}$$

Example: On any given morning, it is either raining (25% of the time), foggy (35% of the time), or sunny (40% of the time). If it is raining, there is a 30% chance that I will miss the bus. If it is foggy, there is a 20% chance that I will miss the bus. If it is sunny, there is a 10% chance that I will miss the bus. What is the probability that I will miss the bus on any given morning?

$$\begin{aligned} P(MB) &= P(MB|R)P(R) + P(MB|F)P(F) + P(MB|S)P(S) \\ &= (0.30)(0.25) + (0.20)(0.35) + (0.10)(0.40) \\ &= 0.075 + 0.07 + 0.04 = 0.185 \end{aligned}$$

Bayesian Analysis

Bayesian analysis is a statistical method that involves using Bayes' theorem to update the probability of a hypothesis based on new evidence or data. It is named after Thomas Bayes, an 18th-century statistician and theologian.

Bayes' Theorem Let $B_1, \dots, B_n$ form a partition of the sample space Ω . Then for any event A :

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_{j=1}^n P(A|B_j)P(B_j)}$$

Note that in the special case of the partition consisting of B and B' , this reduces to:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B')P(B')}$$

Example: If I pull a black marble from one of the boxes in the previous example, what is the probability that I pulled it from Box 1?

$$\begin{aligned} P(B_1|Black) &= \frac{P(Black|B_1)P(B_1)}{P(Black)} \\ &= \frac{\left(\frac{3}{4}\right)\left(\frac{1}{2}\right)}{\frac{5}{8}} = \frac{3}{5} \end{aligned}$$

Example: According to the Arizona Chapter of the American Lung Association, 7% of the population has lung disease. Of those, people, 90% are smokers; and of those without lung disease, 74.7% are non-smokers. What are the chances that a smoker has lung disease?

$$\begin{aligned}
 P(LD|S) &= \frac{P(S|LD)P(LD)}{P(S|LD)P(LD) + P(S|LD')P(LD')} \\
 &= \frac{(0.90)(0.07)}{(0.90)(0.07) + (0.253)(0.93)} \\
 &= \frac{0.063}{0.063 + 0.23529} = \frac{0.063}{0.29829} \approx 0.2111
 \end{aligned}$$

Example: You give a friend a letter to mail. He forgets to mail it with probability 0.2. Given that he mails it, the Post Office delivers it with probability 0.9. Given that the letter was not delivered, what is the probability that it was not mailed.

$$\begin{aligned}
 P(M'|D') &= \frac{P(D'|M')P(M')}{P(D'|M')P(M') + P(D'|M)P(M)} \\
 &= \frac{(1)(0.2)}{(1)(0.2) + (0.1)(0.8)} \\
 &= \frac{0.2}{0.2 + 0.08} = \frac{0.2}{0.28} = \frac{5}{7}
 \end{aligned}$$

2.5 Independence

Definition: Two events A and B are independent if the occurrence of one event does not affect the probability of the other event occurring. Otherwise, the events are dependent. Sometimes it is obvious that two events influence each other, such as crude oil prices and gas prices. Sometimes it is obvious that two events are not dependent, such as rolling a die and the weather. Sometimes it is difficult to tell, such as obtaining two heads in three coin flips and obtaining a head on the first of three flips. Mutual exclusivity and independence are not the same thing. Two events can be independent and not mutually exclusive, such as rolling a 1 and rolling an odd number on a die.

Checking for Independence: In many cases, you will need to evaluate independence by checking the following formulas:

- A and B are independent if and only if $P(A|B) = P(A)$
- A and B are independent if and only if $P(A \cap B) = P(A)P(B)$

Note that independent trials are okay to be assumed to be independent, such as rolling a die or flipping a coin.

Properties:

- If A and B are independent, then b and A are independent.
- If A and B are independent, then A' and B' , A and B' , and A' and B are all independent.

Example: Suppose we draw a card from a deck of 52 playing cards. Let A be the event that the card is an Ace, and let B be the event that the card is a Spade. Are A and B independent?

$$\begin{aligned}P(A) &= \frac{4}{52} = \frac{1}{13} \\P(B) &= \frac{13}{52} = \frac{1}{4} \\P(A \cap B) &= \frac{1}{52} \\P(A)P(B) &= \frac{1}{13} \cdot \frac{1}{4} = \frac{1}{52}\end{aligned}$$

Since $P(A \cap B) = P(A)P(B)$, A and B are independent.

Example: Suppose we roll 2d6. What is the probability of getting 2 even numbers?

$$\begin{aligned}P(E_1 \cap E_2) &= P(E_1)P(E_2) \quad \text{since } E_1 \text{ and } E_2 \text{ are independent} \\&= \left(\frac{3}{6}\right) \left(\frac{3}{6}\right) \\&= \frac{9}{36} = \frac{1}{4}\end{aligned}$$

Example: Roll 2d6. Let A be the event that the sum is 7. Let B be the event that the first die is a 4. Are A and B independent?

$$\begin{aligned}P(A) &= \frac{6}{36} = \frac{1}{6} \\P(B) &= \frac{1}{6} \\P(A \cap B) &= \frac{1}{36} \\P(A)P(B) &= \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}\end{aligned}$$

Since $P(A \cap B) = P(A)P(B)$, A and B are independent.

3. Discrete Random Variables and Probability Distributions

3.1 Random Variables

In many random experiments, we are not interested in the specific outcome of an experiment, but in some characteristic of that outcome. If you play the lottery, for example, you would like the numbers chosen in the weekly drawing to match as many of the number you marked on your ticket as possible. You do not particularly care which of your numbers are matched, but rather how many matches you have.

Definition For a given sample space of some experiment, a **random variable** is any rule that associates a number with each outcome in S . In mathematical language, a random variable is a function whose domain is the sample space S and whose range is the set of real numbers. We usually denote random variables with capital letters such as X , Y , and Z .

$$X : S \rightarrow \mathbb{R}$$
$$\exists s \in S \ni X(s) = x$$

Random Variable Examples

- Sum of two dice
- Number of defects on a randomly selected piece of fabric.
- SAT score of a randomly selected student.
- Number of heads in 10 flips of a coin.
- The amount a company has to pay to an employee in pension.

Types of Random Variables

- A **discrete random variable** is a random variable that can take on a finite or countably infinite number of possible values.
- A **continuous random variable** is a random variable that can take on infinitely many values corresponding to an interval on the real line.

3.2 Probability Distributions of Discrete Random Variables

Probability Distributions

Definition To describe a discrete random variable X , you need two pieces of information:

- What values X can assume.
- The probabilities associated with each of these values.

The **probability distribution** or **probability mass function** (pmf) of a discrete random variable X is a formula, table, or graph that gives both the possible values of X and the probabilities associated with each value of X . The probability distribution or pmf of a discrete random variable is defined for every number x by:

$$p(x) = P(X = x)P(\forall s \in S, X(s) = x)$$

Example: Let X be the random variable that gives the sum of 2d6.

x	2	3	4	5	6	7	8	9	10	11	12
p(x)	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Bernoulli Random Variables

Definition Any random variable whose only possible values are 0 and 1 is called a **Bernoulli random variable**. The PMF looks like this:

x	0	1
p(x)	$1 - p$	p

Example: A box contains one red, two orange, and five black balls. You choose two balls at random. For each red ball you win \$5, for each orange ball you win \$1, and for each black ball you win nothing. Let X be the random variable that gives your total winnings. Find the probability distribution of X and the probability that you will win at most \$3.

x	p(x)	Explanation
0	$\frac{10}{28}$	BB
1	$\frac{10}{28}$	BO, OB
2	$\frac{1}{28}$	OO
5	$\frac{5}{28}$	RB, BR
6	$\frac{2}{28}$	RO, OR

$$P(X \leq 3) = p(0) + p(1) + p(2) = \frac{10}{28} + \frac{10}{28} + \frac{1}{28} = \frac{21}{28} = \frac{3}{4}$$

Cumulative Distribution Function

Definition The **cumulative distribution function** (CDF) $F(x)$ of a discrete RV X with PMF $p(x)$ is defined for **every number** x by:

$$F(x) = P(X \leq x) = \sum_{i \leq x} p(i)$$

Example: Take the previous example with the balls. Some examples of using CDF here are:

$$F(3) = P(X \leq 3) = p(0) + p(1) + p(2) = \frac{21}{28} = \frac{3}{4}$$

$$F(5) = P(X \leq 5) = p(0) + p(1) + p(2) + p(5) = \frac{26}{28} = \frac{13}{14}$$

$$F(-10) = P(X \leq -10) = 0$$

$$F(10) = P(X \leq 10) = 1$$

We can find the CDF:

$$F(x) = \begin{cases} 0 & x < 0 \\ \frac{10}{28} & 0 \leq x < 1 \\ \frac{20}{28} & 1 \leq x < 2 \\ \frac{21}{28} & 2 \leq x < 5 \\ \frac{26}{28} & 5 \leq x < 6 \\ 1 & x \geq 6 \end{cases}$$

Example: Find the PMF from the following CDF:

$$\begin{cases} 0 & x < -3 \\ 0.2 & -3 \leq x < -2 \\ 0.3 & -2 \leq x < 0 \\ 0.6 & 0 \leq x < 1 \\ 0.9 & 1 \leq x < 4 \\ 1 & x \geq 4 \end{cases}$$

PMF:

x	p(x)	Explanation
-3	0.2	$F(-3) - F(-\infty)$
-2	0.1	$F(-2) - F(-3)$
0	0.3	$F(0) - F(-2)$
1	0.3	$F(1) - F(0)$
4	0.1	$F(4) - F(1)$

3.3 Expected Values

The **expected value** is a weighted average of the possible values of a random variable X with PMF $p(x)$. It is given by:

$$E(X) = \mu_X = \sum_x xp(x)$$

where the sum is over all possible values of X .

Names for Expected Value

- Mean
- Expected Value
- Expectation
- μ
- $E(X)$
- (Bonus) The "center of mass" of the histogram of the PMF.

Example: Craps is a dice game in which the players make wagers on the sum of a roll of a pair of dice. One of the wagers available to the player is the *field bet*. The field consists of $\{5, 6, 7, 8\}$. If the sum of the dice is one of these numbers, you lose your bet. Otherwise you get your bet back plus a payout according to the following table:

Winnings	-1	1	2	3
Outcome	field	3, 4, 9, 10, 11	2	12

Is the bet favorable to the player?

Solution: Let X be the random variable that gives your winnings on a field bet. The PMF is given by:

x	-1	1	2	3
$p(x)$	$\frac{16}{36}$	$\frac{20}{36}$	$\frac{1}{36}$	$\frac{1}{36}$

$$\begin{aligned}
 E(X) &= \sum_x xp(x) \\
 &= -1 \cdot \frac{20}{36} + 1 \cdot \frac{14}{36} + 2 \cdot \frac{1}{36} + 3 \cdot \frac{1}{36} \\
 &= -\frac{20}{36} + \frac{14}{36} + \frac{2}{36} + \frac{3}{36} \\
 &= -\frac{1}{36} \\
 &\approx -0.028
 \end{aligned}$$

The expected value is negative, so the bet is unfavorable to the player.

Expectation of a Function of a Random Variable

If X is a discrete random variable with PMF $p(X)$ and $h(x)$ is any real valued function of X , then the expectation of $h(X)$ is given by:

$$E[h(X)] = \sum_x h(x)p(x)$$

Example: Find $E(X^2)$ if X is the sum of two dice.

$$E(X^2) = \sum x^2 p(x)$$

Note that this is not always reversible. $E[h(X)] \neq h(E[X])$ except when $h(x)$ is a linear function. Then

$$E(aX + b) = aE[X] + b$$

Proof:

$$\begin{aligned} E(aX + b) &= \sum_x (ax + b)p(X) \\ &= \sum_x ax \cdot p(x) + b \cdot p(x) \\ &= a \sum_x xp(x) + b \sum_x p(x) \\ &= aE[X] + b \cdot 1 \\ &= aE[X] + b \end{aligned}$$

Special Cases and Rules:

$$\begin{aligned} a = 0 : E(b) &= b \\ b = 0 : E(aX) &= aE[X] \\ E(X + Y) &= E(X) + E(Y) \\ E(3X - 2Y + b) &= 3E(X) - 2E(Y) + b \end{aligned}$$

Variance of a Discrete Random Variable

The **variance** of a discrete random variable is the "spread" of the random variable around its mean. Suppose X is a discrete random variable with PMF $p(x)$ and mean μ . The variance of X , denoted by σ^2 is given by:

$$\sigma^2 = Var(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$$

Recall here that $\mu = E(X)$. The variance is the mean squared distance of the values of X from their mean.

Names for Variance

- Variance
- $Var(X)$
- $V(X)$
- σ^2

Computation Formula

$$V(X) = E[X^2] - (E[X])^2$$

Standard Deviation

The **standard deviation** of a discrete random variable X is the positive square root of the variance of X . It is denoted by σ .

$$\sigma = SD(X) = \sqrt{Var(X)} = \sqrt{E[X^2] - (E[X])^2}$$

Note: Variance is NOT a linear operator. Let X be a random variable with constants a and b . Then:

$$V(aX + b) = a^2V(X) \implies \sigma_{aX+b} = |a|\sigma_X$$

Example: Show that the variance of the Bernoulli Random Variable is $p(1-p)$:

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 \\ &= \sum_x x^2 p(x) - \left(\sum_x x p(x) \right)^2 \\ &= 0^2(1-p) + 1^2(p) - (0(1-p) + 1(p))^2 \\ &= p - p^2 \\ &= p(1-p) \end{aligned}$$

Example: Let X be a discrete random variable with $\mu = 2$ and variance of $\sigma^2 = 6$. (a) Find $E(X^2)$. (b) Find $V[(X - 1)^2]$ and (c) Find σ_X .

$$\begin{aligned} (a) \quad E(X^2) &= V(X) + [E(X)]^2 \\ &= 6 + 2^2 \\ &= 10 \end{aligned}$$

$$\begin{aligned} (b) \quad V[(X - 1)^2] &= E[(X - 1)^4] - [E(X - 1)^2]^2 \\ &= E[X^4 - 4X^3 + 6X^2 - 4X + 1] - [E(X^2) - 2E(X) + 1]^2 \\ &= E[X^4] - 4E[X^3] + 6E[X^2] - 4E[X] + 1 - [10 - 4 + 1]^2 \\ &= E[X^4] - 4E[X^3] + 60 - 8 + 1 - 49 \\ &= E[X^4] - 4E[X^3] + 4 \end{aligned}$$

$$(c) \quad \sigma_X = \sqrt{V(X)} = \sqrt{6} \approx 2.45$$

3.4 The Binomial Probability Distribution

Definition: There are many experiments that conform either exactly or approximately to the following list of requirements:

- The experiment consists of a sequence of n identical trials.
- Each trial results in one of two outcomes, which we will call a success and a failure.
- The trials are independent, so that the outcome on any particular trial does not influence the outcome on any other trial.
- The probability of a success, denoted by p , remains the same from trial to trial. The probability of a failure is $q = 1 - p$.

An experiment that satisfies these requirements is called a **binomial experiment**.

Example: Consider each of the next n vehicles undergoing an emissions test, and let S denote a vehicle that passes the test and F denote one that fails.

- The experiment consists of n identical trials, where each trial is the emissions test of a vehicle.
- Each trial results in one of two outcomes: S or F .
- The trials are independent, assuming that the vehicles are randomly selected.
- The probability of a success (a vehicle passing the test) remains the same from trial to trial, assuming that the vehicles are randomly selected.

Thus, this is a binomial experiment.

The Binomial Random Variable and Distribution

In most binomial experiments, we are interested in the total number of successes that occur in the n trials rather than the specific sequence of successes and failures.

Definition: The **binomial random variable** X associated with a binomial experiment consisting of n trials is defined as:

X = the number of successes in n trials.

Example: Suppose that $n = 3$. There are 8 possible outcomes of the experiment, as shown in the following table:

Outcome	X
SSS	3
SSF	2
SFS	2
FSS	2
SFF	1
FSF	1
FFS	1
FFF	0

The possible values of X are 0, 1, 2, and 3.

Example: Consider the case that $n = 4$. Find $P(SSFS)$ and $P(SSSS)$:

$$\begin{aligned}P(SSFS) &= P(S) \cdot P(S) \cdot P(F) \cdot P(S) \\&= p \cdot p \cdot (1 - p) \cdot p \\&= p^3(1 - p) \\P(SSSS) &= P(S) \cdot P(S) \cdot P(S) \cdot P(S) \\&= p \cdot p \cdot p \cdot p \\&= p^4\end{aligned}$$

Notation: Because the pmf of a binomial random variable X depends on the two parameters n and p , we denote the pmf by $b(x : n, p)$.

Example: Find $b(3; 4, p)$:

$$\begin{aligned} b(3; 4, p) &= P(X = 3) \\ &= P(SSSF) + P(SSFS) + P(SFSS) + P(FSSS) \\ &= p^3(1 - p) + p^3(1 - p) + p^3(1 - p) + p^3(1 - p) \\ &= 4p^3(1 - p) \end{aligned}$$

Theorem :

$$b(x; n, p) = \begin{cases} \binom{n}{x} p^x (1 - p)^{n-x}, & x = 0, 1, 2, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

Example: Each of six randomly selected cola drinkers is given a glass containing cola S and one containing cola F . The glasses are identical in appearance except for a code on the bottom to identify the cola. Suppose there is actually no tendency among cola drinkers to prefer one cola over the other. Then $p = P(S) = 0.5$. Find (1) the probability that exactly four of the six cola drinkers prefer cola S , (2) the probability that at least four of the six cola drinkers prefer cola S , and (3) the probability that at most 1 of the six cola drinkers prefer cola S .

1.

$$\begin{aligned} P(X = 4) &= b(4; 6, 0.5) = \binom{6}{4} (0.5)^4 (1 - 0.5)^{6-4} \\ &= \binom{6}{4} (0.5)^4 (0.5)^2 \\ &= 15 \cdot (0.5)^6 \\ &= \frac{15}{64} \approx 0.2344 \end{aligned}$$

2.

$$\begin{aligned}P(X \geq 4) &= \sum_{x=4}^6 b(x; 6, 0.5) \\&= \sum_{x=4}^6 \binom{6}{x} (0.5)^x (1 - 0.5)^{6-x} \\&= \frac{22}{64} \approx 0.3438\end{aligned}$$

3.

$$\begin{aligned}P(X \leq 1) &= \sum_{x=0}^1 b(x; 6, 0.5) \\&= \sum_{x=0}^1 \binom{6}{x} (0.5)^x (1 - 0.5)^{6-x} \\&= \frac{7}{64} \approx 0.1094\end{aligned}$$

Binomial Tables

For $X \sim \text{Binomial}(n, p)$, the cdf will be denoted by:

$$B(x; n, p) = P(X \leq x) = \sum_{y=0}^{\lfloor x \rfloor} b(y; n, p)$$

Theorem: If $X \sim B(n, p)$, then:

- $E(X) = np$
- $V(X) = np(1 - p)$
- $\sigma_X = \sqrt{np(1 - p)}$

3.5 Hypergeometric and Negative Binomial Probability Distributions

Negative Binomial Distribution

Definition: A negative binomial experiment is a sequence of independent and identical Bernoulli trials that continues until the r^{th} success is observed. The negative binomial random variable X is defined as the number of trials required to obtain r successes, where r is a positive integer. This is denoted:

$$X \sim \text{Negative Binomial}(r, p) \equiv X \sim NB(r, p)$$

Examples:

- The number of times you flip a fair coin until you observe heads for the fourth time.
- The number of flies you catch until you have caught 10 that are female.
- The number of customers that enter a bank until 5 of them make a deposit.

Example: Suppose X = the number of times you have to flip a coin until you obtain the three heads. Find $P(X = 5)$.

$$\begin{aligned} P(X = 5) &= \binom{5-1}{3-1} \left(\frac{1}{2}\right)^3 \left(1 - \frac{1}{2}\right)^{5-3} \\ &= \binom{4}{2} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 \\ &= 6 \left(\frac{1}{2}\right)^5 = \frac{6}{32} = \frac{3}{16} \end{aligned}$$

PMF: The possible values of a Negative Binomial random variable X are $x = r, r + 1, r + 2, \dots$. The probability that the r^{th} success occurs on the k^{th} trial is given by the pmf:

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \quad k = r, r + 1, r + 2, \dots$$

Note: If $X_1, \dots, X_r$ are independent geometric random variables with parameter p , then:

$$X = \sum_{i=1}^r X_i \sim NB(r, p)$$

Mean and Variance: The expected value and variance of a Negative Binomial random variable with parameters r and p are given by:

$$E(X) = \frac{r}{p}$$

$$V(X) = \frac{r(1-p)}{p^2}$$

Example: You roll a fair six-sided die until you see the third six. Define the random variable you are working with in words and state the corresponding distribution along with all relevant parameters. (a) What is the probability that this happens on the tenth roll? (b) How many rolls do you expect to make?

Let $X \sim NB(3, \frac{1}{6})$ be the number of rolls until the third six.

a.

$$\begin{aligned} P(X = 10) &= \binom{10-1}{3-1} \left(\frac{1}{6}\right)^3 \left(1 - \frac{1}{6}\right)^{10-3} \\ &= \binom{9}{2} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^7 \\ &= 36 \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^7 \\ &= 36 \cdot \frac{1}{216} \cdot \frac{78125}{279936} \\ &= \frac{36 \cdot 78125}{216 \cdot 279936} \\ &\approx 0.0845 \end{aligned}$$

b.

$$E(X) = \frac{r}{p} = \frac{3}{\frac{1}{6}} = 18$$

Geometric Probability Distribution

Definition: This is a special case of the negative binomial distribution where $r = 1$. Essentially we are looking for the number of trials required to obtain the first success. This is denoted:

$$X \sim \text{Geometric}(p)$$

The pmf is given by:

$$P(X = k) = p(1 - p)^{k-1}, \quad k = 1, 2, 3, \dots$$

The mean and variance are given by:

$$E(X) = \frac{1}{p}, \quad V(X) = \frac{1 - p}{p^2}$$

Review of Geometric Series:

$$1 + A + A^2 + A^3 + \dots + A^n = \begin{cases} \frac{1 - A^{n+1}}{1 - A}, & A \neq 1 \\ n + 1, & A = 1 \end{cases} \quad \frac{1 - A^{n+1}}{1 - A}$$

$$\sum_{k=0}^{\infty} A^k = \frac{1}{1 - A}, \quad |A| < 1$$

Example: You roll a fair die until you see the first six. Define the random variable you are working with in words and state the corresponding distribution along with all relevant parameters. (a) How many rolls do you expect to make? (b) What is the probability that the first six occurs on the fifth roll? (c) What is the probability that you will need more than 10 rolls?

Let $X \sim \text{Geo}(\frac{1}{6})$ be the number of rolls until the first six.

a.

$$E(X) = \frac{1}{p} = \frac{1}{\frac{1}{6}} = 6$$

b.

$$\begin{aligned}P(X = 5) &= p(1 - p)^{5-1} \\&= \frac{1}{6} \left(1 - \frac{1}{6}\right)^4 \\&= \frac{1}{6} \left(\frac{5}{6}\right)^4 \\&= \frac{625}{7776} \approx 0.0804\end{aligned}$$

c.

$$\begin{aligned}P(X > 10) &= 1 - P(X \leq 10) \\&= 1 - \sum_{k=1}^{10} P(X = k) \\&= 1 - \sum_{k=1}^{10} p(1 - p)^{k-1} \\&= 1 - p \sum_{k=0}^9 (1 - p)^k \\&= 1 - p \left(\frac{1 - (1 - p)^{10}}{1 - (1 - p)} \right) = 1 - (1 - (1 - p)^{10}) \\&= (1 - p)^{10} = \left(1 - \frac{1}{6}\right)^{10} = \left(\frac{5}{6}\right)^{10} \approx 0.1615\end{aligned}$$

Remark: The textbook uses Y , the number of failures before the first success, instead of X , the number of trials until the first success. This means that $Y = X - 1$.

Hypergeometric Probability Distribution

Definition: Suppose you have a population of size N , in which each individual can be classified as either a success (S) or a failure (F). The number of successes in the population is M . A sample of size n is selected at random without replacement from the population. Let X be the number of successes

in the sample. Then X is a hypergeometric random variable with parameters n , M , and N :

$$X \sim \text{Hypergeometric}(n, M, N) \equiv X \sim HG(n, M, N)$$

The pmf is denoted:

$$h(n, M, N)$$

Example: Identify the parameters n , M , and N for the following situations:

- The number of aces in a poker hand.
 - $N = 52$ (cards in deck)
 - $M = 4$ (aces in deck)
 - $n = 5$ (cards in hand)
- The number of juniors in a 5-person committee chosen at random from a group of ten juniors and five seniors.
 - $N = 15$ (total students)
 - $M = 10$ (juniors)
 - $n = 5$ (committee members)
- The number of green marbles you pull when randomly selecting 3 marbles from a bag with 10 red, 8 green, and 2 blue marbles.
 - $N = 20$ (total marbles)
 - $M = 8$ (green marbles)
 - $n = 3$ (marbles selected)

Let X = the number of green marbles you pull when randomly selecting 3 marbles from a bag with 10 red, eight green, and two blue marbles. Find

the PMF of X .

$$\begin{aligned}
 P(X=0) &= \frac{\binom{8}{0}\binom{12}{3}}{\binom{20}{3}} = \frac{220}{1140} = \frac{22}{114} \approx 0.19298 \\
 P(X=1) &= \frac{\binom{8}{1}\binom{12}{2}}{\binom{20}{3}} = \frac{8 \cdot 66}{1140} = \frac{528}{1140} = \frac{44}{95} \approx 0.46316 \\
 P(X=2) &= \frac{\binom{8}{2}\binom{12}{1}}{\binom{20}{3}} = \frac{28 \cdot 12}{1140} = \frac{336}{1140} = \frac{28}{95} \approx 0.29474 \\
 P(X=3) &= \frac{\binom{8}{3}\binom{12}{0}}{\binom{20}{3}} = \frac{56 \cdot 1}{1140} = \frac{56}{1140} = \frac{14}{285} \approx 0.04912
 \end{aligned}$$

PMF: Generically, the pmf is given by:

$$h(x; n, M, N) = P(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}, \quad \max\{n-(N-M), 0\} \leq x \leq \min\{n, M\}$$

$$\begin{aligned}
 E(X) &= n \left(\frac{M}{N} \right) \\
 V(X) &= n \left(\frac{M}{N} \right) \left(\frac{N-M}{N} \right) \left(\frac{N-n}{N-1} \right)
 \end{aligned}$$

Example: Let X = number of aces in a 5 card poker hand. (a) Describe the distribution of X . (b) What is the probability that $x = 2$? (c) How many aces do you expect to have?

a.

$$X \sim HG(5, 4, 52)$$

b.

$$\begin{aligned}
 P(X=2) &= h(2; 5, 4, 52) \\
 &= \frac{\binom{4}{2}\binom{48}{3}}{\binom{52}{5}} \\
 &= \frac{6 \cdot 17296}{2598960} \\
 &= \frac{103776}{2598960} \approx 0.03993
 \end{aligned}$$

c.

$$E(X) = 5 \left(\frac{4}{52} \right) = \frac{5}{13} \approx 0.3846$$

Relationship Between the Binomial and Hypergeometric Distributions

Remark: The hypergeometric distribution can be approximated by the binomial distribution when the population size N and the number of successes M is large relative to the sample size n .

3.6 Poisson Probability Distribution

Definition: The **Poisson distribution** is used to model the number of times a rare event occurs.

Examples:

- The number of earthquakes in CA in a month.
- The number of misprints on a page of some document.
- The number of atoms emitted by a radioactive particle.

A random variable X is called a Poisson (λ) random variable ($\lambda > 0$), if X takes on values $0, 1, 2, \dots$ and has pmf:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

- The expected value $E(X) = \lambda$
- The variance $V(X) = \lambda$
- The parameter λ is the average number of occurrences in a given interval.

Note: Informally, the smaller λ is, the more rare the event X is.

Example: Suppose on average there are two typos per page in the text book.

- For a randomly selected page of the book, what is the probability that there are no typos?
- For a randomly selected page of the book, what is the probability that there are two typos?
- For a randomly selected page of the book, what is the probability that there is at least 1 typo?

Let $X \sim \text{Poisson}(2)$ be the number of typos on a randomly selected page.

- $\lambda = 2$
- $P(X = 0) = \frac{e^{-2}2^0}{0!} = e^{-2} \approx 0.1353$
- $P(X = 2) = \frac{e^{-2}2^2}{2!} = 2e^{-2} \approx 0.2707$
- $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-2} \approx 0.8647$

Example: Both Hurricane Harvey and Hurricane Irma were said to be "One in 500 Year" storms, yet they both occurred in 2017.

1. What is the probability that 2 such storms occur in a given year? Same month?
2. That is the probability that a baby born today in a hurricane zone will see one such storm. What about a "one in 100 year" storm?

Let $X \sim \text{Poisson}(\lambda)$ be the number of "one in 500 year" storms in a year and $Y \sim \text{Poisson}(\mu)$ be the number of "one in 500 year" storms in a month.

- 1.

$$\begin{aligned}\lambda &= \frac{1}{500} \\ P(X = 2) &= \frac{e^{-\frac{1}{500}} \left(\frac{1}{500}\right)^2}{2!} \\ &\approx 1.996 \times 10^{-6}\end{aligned}$$

$$\begin{aligned}\mu &= \frac{1}{500 \cdot 12} = \frac{1}{6000} \\ P(Y = 2) &= \frac{e^{-\frac{1}{6000}} \left(\frac{1}{6000}\right)^2}{2!}\end{aligned}$$

2.

$$\begin{aligned}\lambda &= \frac{1}{100} \\ P(X \geq 1) &= 1 - P(X = 0) = 1 - e^{-\frac{1}{100}} \\ &\approx 0.00995\end{aligned}$$

$$\begin{aligned}\mu &= \frac{1}{100 \cdot 12} = \frac{1}{1200} \\ P(Y \geq 1) &= 1 - P(Y = 0) = 1 - e^{-\frac{1}{1200}} \\ &\approx 0.000833\end{aligned}$$